Sabarishwaran G

sabarishwaran.ganesan@gmail.com — +91 8610605084 — LinkedIn — Portfolio — Trichy, Tamil Nadu

Skills

Languages & Tools: Python, SQL (MySQL, PostgreSQL, BigQuery), Cypher, Bash, Git, REST APIs, Flask
Generative & Agentic AI: LLMs (Claude, AWS Bedrock), Multi-Agent Systems, LangGraph, LangChain, Tool Calling, RAG, GraphRAG, Vector Search, Human-in-the-Loop, LLM-as-Judge, Prompt Engineering
Machine Learning & AI: Deep Learning, Transformers, Autoencoders, NLP, Computer Vision, Reinforcement Learning, LightGBM, XGBoost, PyTorch, Scikit-learn, SHAP, Statistical Modeling, A/B Testing
Data & Visualization: PySpark, Apache Arrow, EDA, Feature Engineering, Matplotlib, Seaborn, Plotly
Cloud & MLOps: AWS (SageMaker, Lambda, EC2, S3, EMR, Athena), GCP (Vertex AI Pipelines, BigQuery, Cloud Run, GCS, Cloud Scheduler, IAM), Kubeflow Pipelines (KFP), Terraform, Docker

Work Experience

Data Scientist — Trellix *July 2023 – Present*

- **STREAM: Multi-Agent LLM System for Security Escalation Automation**
  - Built an agentic system that ingests customer security escalations from Jira, triages and investigates them with LLM agents, and proposes resolutions or hands off to analysts via **human-in-the-loop** review.
  - Designed a **LangGraph orchestrator agent** (Claude) that delegates to product-specific **sub-agents**.
  - Implemented durable HITL via LangGraph interrupts and Postgres checkpoints, resuming on analyst reply.
  - Enforced guardrails: read-only agent tools, deterministic Jira actions, confidence gating, LLM cost budgets.
  - Built hybrid triage (regex + LLM fallback) and dedup via vector search, RRF, and an **LLM-as-judge**.
  - Modeled tickets and indicators as a **Neo4j knowledge graph** with vector search for GraphRAG context.
  - **Technologies:** Python, LangGraph, LangChain, Claude, Bedrock, Neo4j, PostgreSQL, LangSmith, GCP
- **ML-Powered Threat Detection for EDR Systems**
  - Replaced high-volume, manually managed YARA rules with a supervised ML model classifying EDR telemetry into **353 rule types**, achieving **94.5% weighted F1**.
  - Built a feature pipeline: raw telemetry → behavioral sentences → **384-dim transformer embeddings**.
  - Improved detection quality via targeted retraining on misclassified samples, boosting **57 rules to 90% precision**; integrated an LLM to generate detection insights and MITRE ATT&CK mappings.
  - **Technologies:** Python, LightGBM, Scikit-learn, Transformers, AWS (EC2, SageMaker, S3)
- **End-to-End MLOps Pipeline for Malware Classification (GCP)**
  - Owned a production 7-stage ML pipeline (inventory → labeling → extraction → sampling → HP tuning → training → evaluation) on Vertex AI Pipelines and Cloud Run Jobs, classifying Windows PE files over **2,598+ extracted features**.
  - Resolved conflicting labels across **6 upstream sources** (VirusTotal, ReversingLabs, AVTest, DoD, endpoint feedback, manual analysts) via a tiered combiner with versioned label logic; chronological train/val/test splits prevent temporal leakage.
  - Trained LightGBM on Vertex AI GPU CustomJobs with a 3-tier hyperparameter fallback (Bayesian HP-tuning winner → config defaults → library defaults); tuned 8 params with parallel Bayesian trials.
  - Scaled to **120M+ row** training sets via Arrow streaming over the BigQuery Storage Read API, dtype compaction (f64→f32, i64→i32), and Parquet split caching on GCS.
  - Automated infrastructure with Terraform (BQ datasets, GCS buckets, IAM, daily Cloud Scheduler retrains) and decoupled config from images via GCS-hydrated YAML.
  - **Technologies:** Python, LightGBM (GPU/OpenCL), GCP (Vertex AI, BigQuery, Cloud Run, GCS, Cloud Scheduler, IAM), Kubeflow Pipelines, Terraform, Docker, Apache Arrow
- **Alert Prioritization & Threat Detection**
  - Designed and implemented an AutoEncoder-based alert prioritization system after extensive research into industry best practices.
  - Reduced false positives from **70% to 30%** and cut alert fatigue by **94.5%**, improving SOC efficiency.
  - **Technologies:** Python, PyTorch, Scikit-learn, PySpark, Git, AWS (EC2, SageMaker, S3, EMR, Athena)

Data Science Intern — Trellix *Mar 2023 – Jun 2023*

- **URL Categorization using Machine Learning**
  - Built an NLP-based URL categorization pipeline with multilingual **XLM transformer** embeddings and **XGBoost**, achieving **76% accuracy** across 60 categories; used **SHAP** for interpretability.

Education

B.Tech in Computer Science Engineering (Artificial Intelligence) *Sept 2019 – May 2023*
Amrita Vishwa Vidyapeetham, Coimbatore
CGPA: 8.02 / 10